Architecture Selection Justification

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Problem Statement: **Household Carbon Footprint & Target Tracker System**

We chose Layered Architecture for the Household Carbon Footprint & Target Tracker System.

Architectural choice

The system is organised into three horizontal layers: a Presentation Layer (User Interface Component), a Business Layer (Log Ingestion, Emission Calculator, and Milestone Tracker Components), and a Data Layer (Data Storage Component). Each layer only communicates with the layer directly beneath it, which keeps the ingestion-to-calculation-to-tracking workflow easy to trace end to end.

Two scenario-related reasons

1. **Workflow fit:** The core functionality — a Resident logs monthly electricity/gas/commute data, the system computes CO2e, and results feed a milestone chart for the Resident and Sustainability Coach — is a single, well-defined pipeline (log → compute → track). A layered structure mirrors this pipeline directly, so each component maps cleanly to one stage of the workflow.
2. **Right-sized complexity:** The system has two actors and a moderate, predictable load (monthly log entries, periodic chart views), not the independently-scaling, high-traffic services that justify a Microservices architecture. Layered architecture gives the needed separation of concerns for testing and maintenance without the operational overhead (service discovery, network calls, distributed data consistency) that Microservices would add.

Security advantage

Because every request must pass through the Business Layer before it can reach the Data Layer, access control and input validation (for example, restricting which aggregate household data a Sustainability Coach can view, or rejecting zero/negative emission values per FR-001's acceptance criteria) can be enforced at one central choke point rather than duplicated across many independent services.

Performance benefit

Since the Emission Calculator and Milestone Tracker Components call the Data Storage Component through in-process, in-layer calls rather than network hops between separately deployed services, the layered design avoids the added network latency of a Microservices approach — helping the historical comparison chart in NFR-001 meet its under-200ms render target.

Component Diagram:

